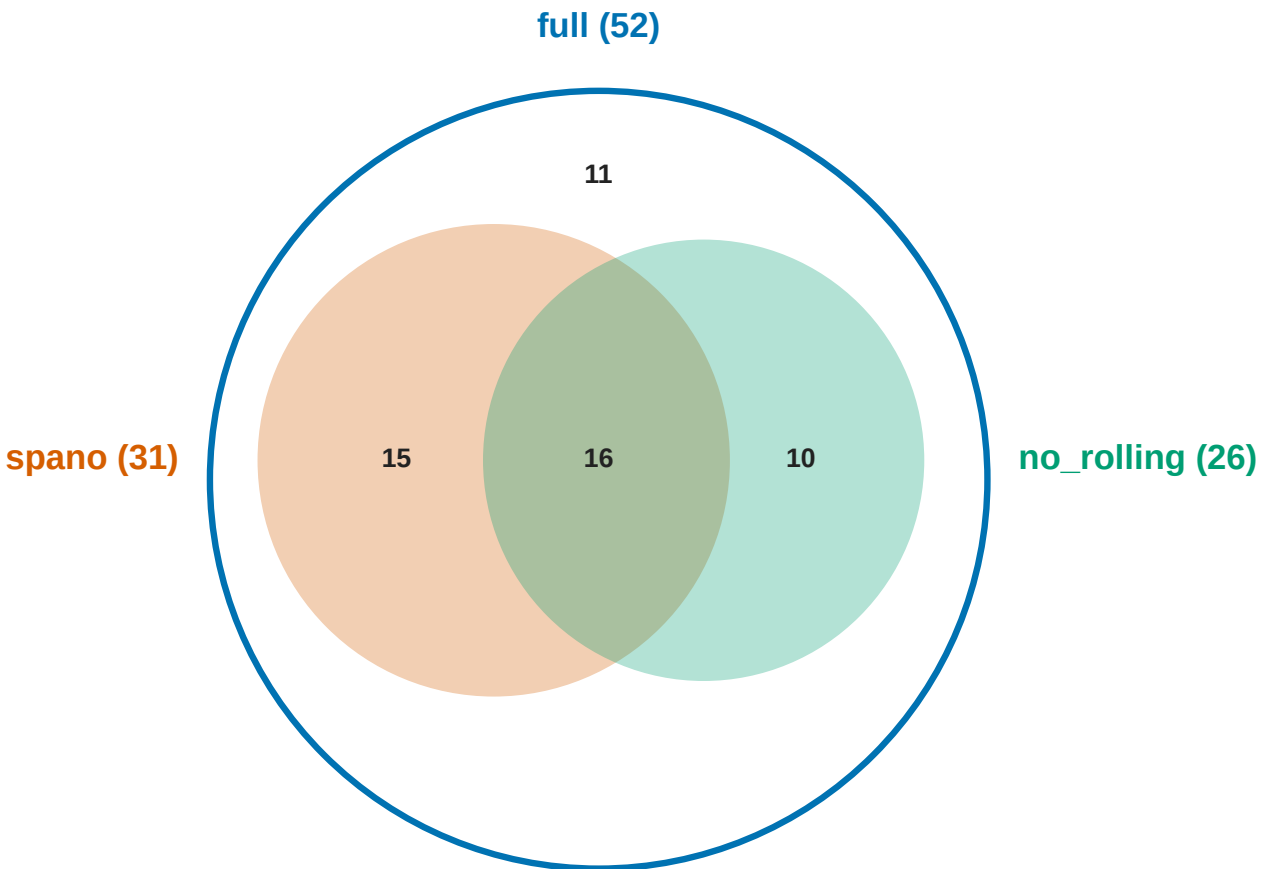


Feature-set inclusion - every feature name, colour-coded by origin

a

Feature-set structure - nested Euler (schematic)



b

All feature names by region (colour = origin)

- full n spano n no_rolling (16)**

 - alcohol_today
 - any_sleep_issue_today
 - cheese_chocolate_today
 - dow
 - excessive_caffeine_today
 - excessive_smoking_today
 - exercise_today
 - irregular_meals_today
 - lack_of_sleep_today
 - menstruation_today
 - overeating_today
 - oversleeping_today
 - ovulation_today
 - stress_today
 - travel_today
 - weather_change_today
- full n spano (not no_rolling) (15)**

 - consecutive_exercise_days
 - consecutive_sedentary_days
 - consecutive_stress_days
 - consecutive_trigger_days
 - consecutive_weather_changes
 - days_since_last_migraine
 - exercise_days_7day
 - migraine_rate_last3
 - migraine_rate_last7
 - migraine_yesterday
 - recent_weekend_sleep_issues
 - sleep_debt_3day
 - sleep_variability_7day
 - stress_drop_today
 - weather_instability_3day
- full only (11)**

 - cheese_chocolate
 - days_since_last_record
 - excessive_smoking
 - exercise_as_trigger
 - headache_free_streak
 - inappropriate_lighting
 - recording_gap_flag
 - sleep_disruption_today
 - sunlight
 - weather_change_yesterday
 - weather_headache_interaction
- full n no_rolling (not spano) (10)**

 - emotional_changes_today
 - exercise_as_trigger_today
 - inappropriate_lighting_today
 - moderate_exercise_min
 - no_exercise_today
 - noise_today
 - physical_fatigue_today
 - specific_smells_today
 - sunlight_today
 - vigorous_exercise_min
- Park (2016) [Tab. 4] subset (6)**

 - alcohol_today
 - hormonal_changes_today
 - noise_today
 - overeating_today
 - stress_today
 - travel_today

● original SHD column or 1:1 rename ● engineered: rolling / lag / interaction / state-derived